

SOFTWARE ENGINEERING PROJECT

E-commerce platform

Prepared by

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Description

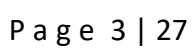
The e-commerce platform will be a web-based application that is accessible to both customers and administrators. It will serve as an online marketplace for devices and support device product listing and management. Customers will be able to browse the catalog, add products to their shopping carts and proceed to checkout. Administrators will have the ability to add, edit, and delete device product listings. The platform will process orders, including secure payment transactions and generating order confirmations. It will also enable order tracking, allowing customers to monitor the status of their shipments in real time. The primary goal of the platform is to improve the user experience, provide intuitive navigation, secure transactions, and efficient order management tailored to devices.

- Enter the e-commerce platform.
- The customer must have a registered account on the e-commerce platform. If he doesn't have an account, he registers a new one via email or through his social media credentials.
- Then the customer logs in to the platform.
- The customer browses the hardware product catalog, searches for specific devices, and views information about the product including price, description, and availability.
- The customer adds the required devices to his shopping cart.
- The customer reviews the shopping cart and makes any necessary adjustments to quantities or removal of devices.
- The customer enters shipping and billing information.
- The customer chooses the preferred payment method (e.g. credit/debit card, PayPal).
- The customer reviews the order summary and confirms the purchase.
- The platform processes the payment and generates the order confirmation.
- Administrators receive the order details and begin the shipping process.
- The customer receives notifications and updates about the order status, including tracking information.
- Customer can track the status of their orders in real time using the tracking information provided.
- The ordered devices are delivered to the customer's specified shipping address.
- The customer receives the devices and is satisfied with his purchase.

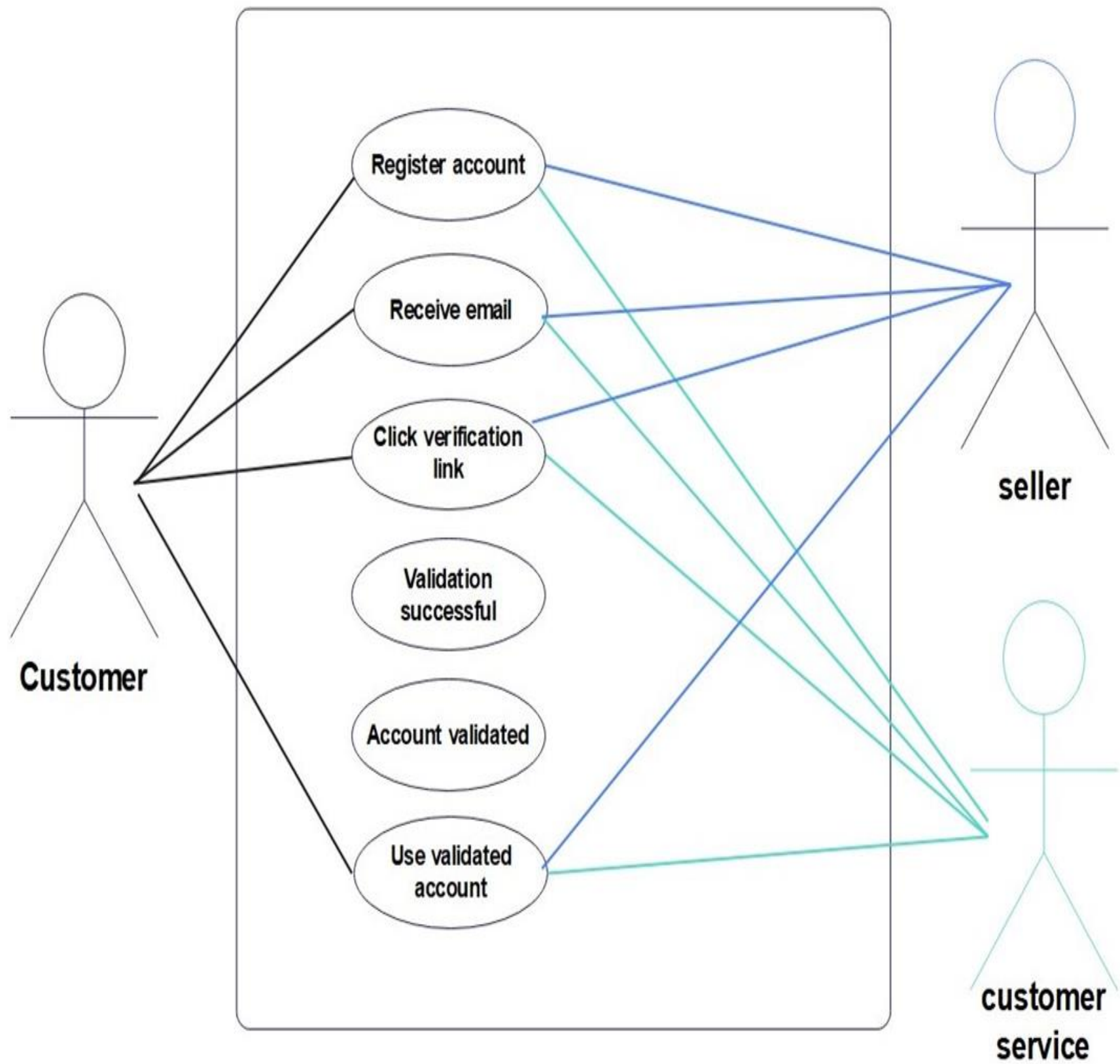
If a payment transaction fails, customers are notified and provided with alternative payment options or assistance to complete the payment process. If a device product becomes out of stock before a customer completes checkout, the customer will be notified and given the option to remove the product from their cart, choose a different device, or join a waitlist to be notified when the product is back in stock.

If the customer wishes to cancel the order before it is shipped, the platform allows him to start the cancellation process. Officials are notified, the order is canceled and the money is returned to the customer if possible. If a customer receives a defective or unsatisfactory device, he or she can submit a return request. The platform provides a simplified process for returning the device and initiating a refund or replacement. If customers encounter problems or have inquiries, they can access customer service through the platform. Customer service team provides assistance and resolves any concerns immediately.

a. Draw the use case model that satisfies the requirements listed in the project topics file with a documentation of each use case.



Scenario Validate Account Use Case Model



Use case name: Validate Account

Summary: This use case describes the process of validating a customer's, seller and customer service account on an e-commerce platform.

Actors:

1. Customer
2. seller
3. customer service

Preconditions:

1. The customer, seller and customer service have created an account on the ecommerce platform.

Main Sequence:

1. The customer, seller and customer service register an account on the e-commerce platform by providing required information such as name email address, and password.
2. After registration, the e-commerce platform sends a verification email to the customer's, seller and customer service provided email address.
3. The customer checks their email and finds the verification email.
4. The customer, seller and customer service open the verification email and clicks on the verification link provided within the email.
5. The e-commerce platform receives the verification request and validates the link.
6. If the verification link is valid, the e-commerce platform confirms the account validation.
7. The customer's, seller and customer service accounts is now considered validated and is ready for use on the e-commerce platform.

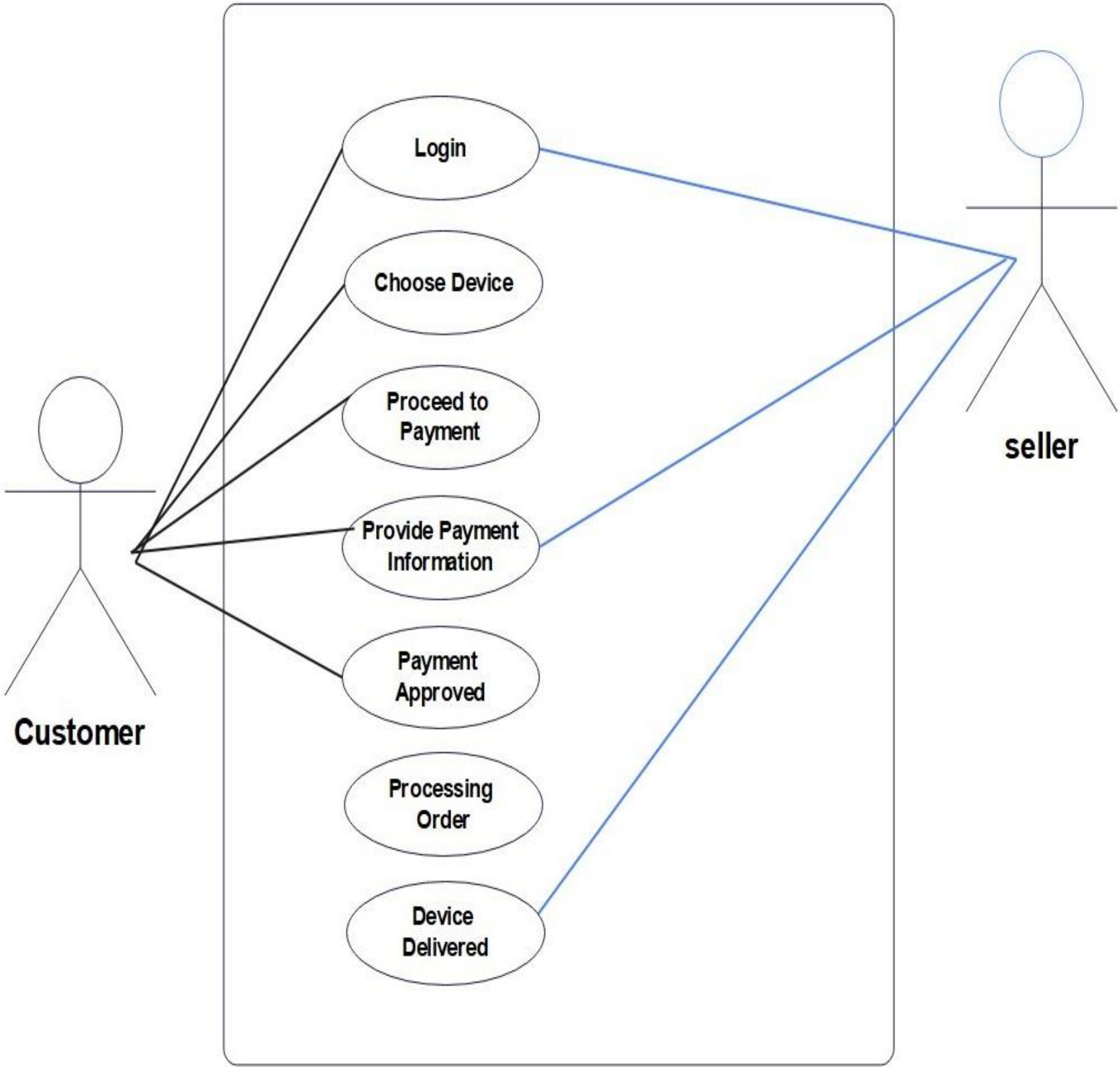
Alternative Sequence:

- If the customer, seller and customer service does not receive the verification email within a reasonable time, they can request the e-commerce.
- platform to resend the verification email.
- In some cases, instead of clicking on a verification link, the customer, seller and customer service may be required to enter a verification code.
- provided in the email on the e-commerce platform's website.

Postcondition:

1. The customer's, seller and customer service account is successfully validated.

Scenario Purchase the device Use
Case Model



Use case name: Purchase the device

Summary: This use case describes the process of purchasing a device on an e-commerce platform.

Actors:

1. Customer
2. Seller

Dependency: Validate account use case.

Preconditions:

1. The customer must have an account on the e-commerce platform.
2. The customer is logged into his account.

Main sequence:

1. The customer logs in to his account on the e-commerce platform.
2. The customer browses the devices available on the e-commerce platform.
3. The customer chooses the device he wants to buy.
4. The customer adds the device to his shopping cart.
5. The customer continues the payment process.
6. The e-commerce platform prompts the customer to review the device details, shipping address, and payment information.
7. The customer provides the necessary payment details, such as credit card information or chooses an alternative payment method.
8. The e-commerce platform communicates with the payment gateway to process the payment transaction securely.
9. The payment gateway verifies the validity of the payment information and verifies that there are sufficient funds or available balance on the customer's chosen payment method.
10. If the payment is approved and successful, the payment gateway sends a confirmation to the ecommerce platform.
11. The e-commerce platform generates an order confirmation that includes details such as order number, device purchased, billing information, shipping details, and expected delivery date.
12. The e-commerce platform begins processing the order, prepares the device for shipping, and provides the customer with tracking information if available.
13. The device is delivered to the customer's specified shipping address within the estimated delivery time frame.

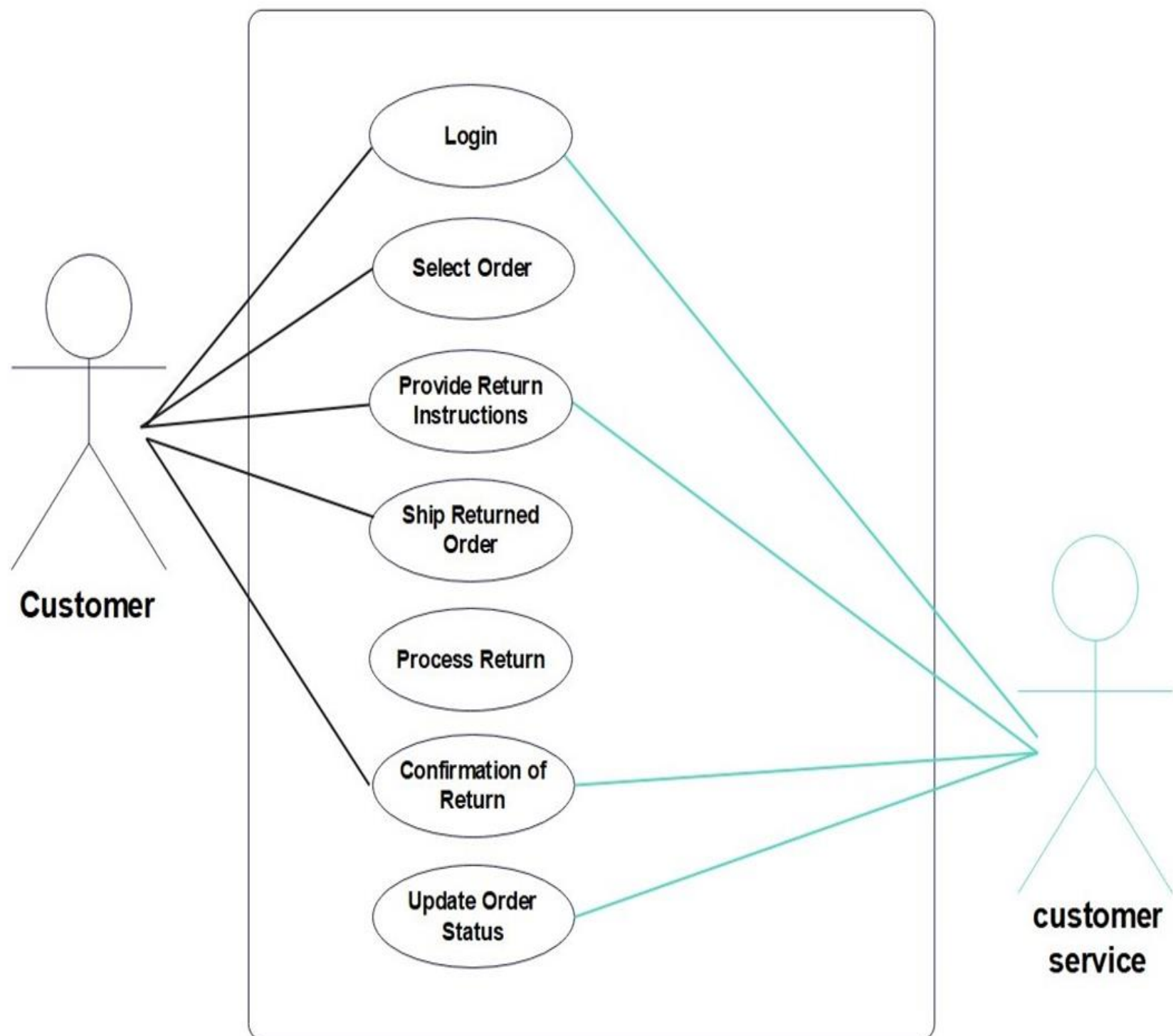
Alternate sequence:

- If a customer encounters a problem during the payment process, he can contact customer support for assistance.
- If the customer decides to cancel the purchase before confirming the order, he can remove the device from the shopping cart or return to continue browsing.
- If a payment transaction fails, the customer is notified and can retry using a different payment method or contact customer support for assistance.

postcondition:

1. The customer's device purchase has been successfully processed.

Scenario Return order Use Case Model



Use case name: Return order

Summary: Filter This use case returns the order to the e-commerce platform.

Actors:

1. Customer
2. customer service.

Dependency: Validate account use case.

Precondition:

1. The customer has received the product from the seller.
2. The customer wants to initiate a return within the specified return period.
3. The customer must have an active account on the e-commerce platform.

Main sequence:

1. He must log in to his account on the e-commerce platform.
2. The customer goes to the order history page or order details page.
3. The customer selects the order he wants to return.
4. The e-commerce platform offers returns for orders, such as refund or exchange.
5. The customer chooses the return option according to the reason for the return.
6. The e-commerce platform sends a return request, and the customer provides return instructions, including any required return shipping labels or documentation.
7. After that, the customer requests a return and follows the following steps.
8. Customer will ship the returned order using the customization method for any required return shipping labels or documentation.
9. Receiving the electronic request for the returned trade platform.
10. The e-commerce platform checks the status of the returned order and processes the return accordingly.
11. If the return is accepted, the e-commerce platform will initiate the refund process or issue instructions regarding the return option chosen by the customer.
12. The customer must notify or confirm the positive status of applicable returns or exchanges
13. Trigger the e-commerce platform and order status to reflect the return and refund /exchange process.

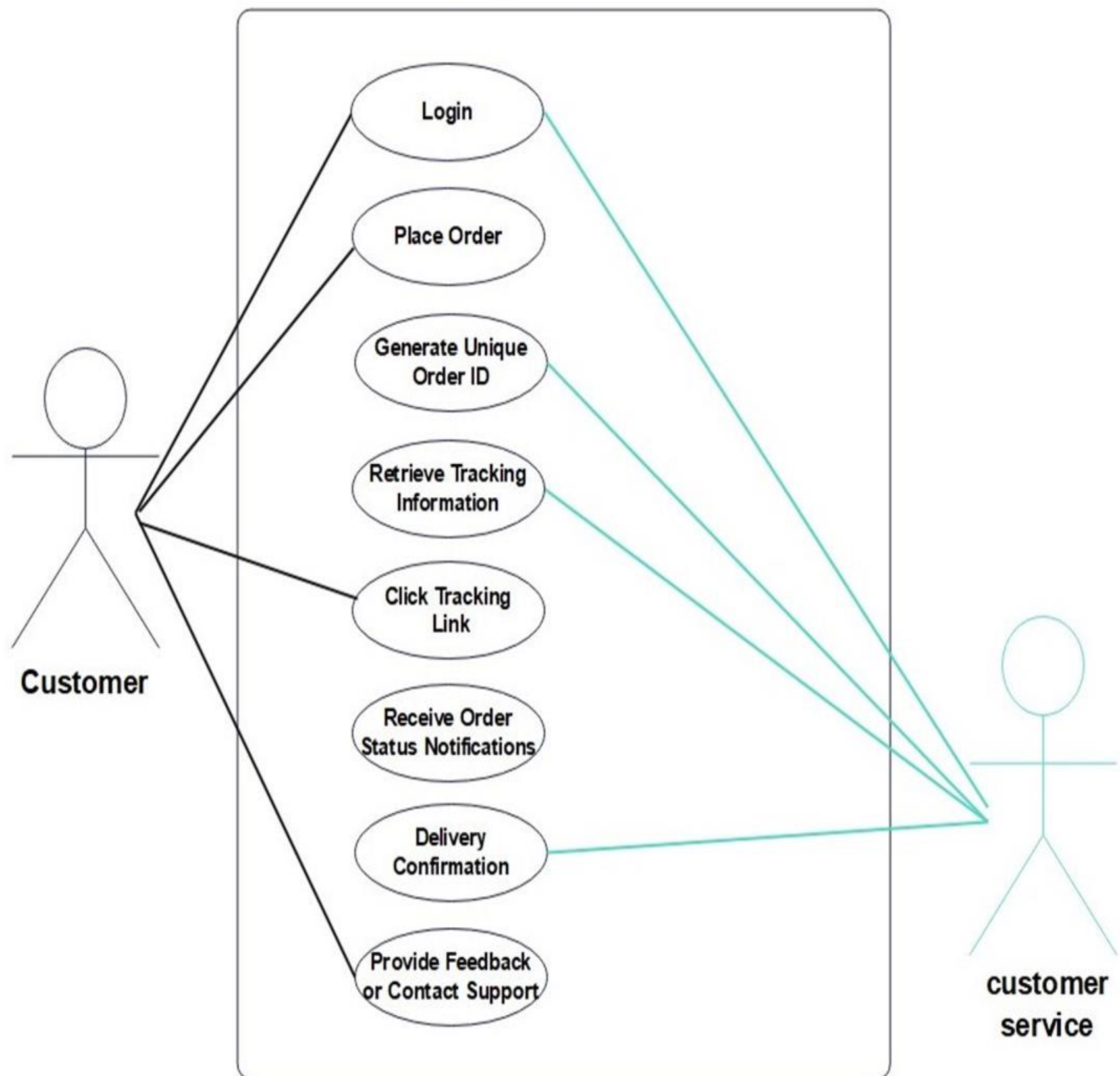
Alternative sequence:

- If the customer receives anything during the response, he can contact the customer to provide assistance.
- If the return request is rejected, the e-commerce platform will explain the reason and may provide solutions or alternative options for the request.
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postcondition:

1. The returned order is received and verified through the e-commerce platform.
2. Refunds are initiated or fully executed based on the return option selected by the customer.
3. The order status is updated for the return and refund/exchange process.

Scenario Order Tracking Use Case Model



Use case name: Order Tracking

Summary: This use case describes the process of with order tracking information on an e-commerce platform.

Actors:

1. Customer.
2. customer service.

Dependency: Validate account use case.

Precondition:

1. The customer has placed an order on the e-commerce platform.
2. The e-commerce platform has integrated order tracking functionality.
3. The customer service has partnered with the e-commerce platform and provides tracking information.

Main Sequence:

1. The customer places an order on the e-commerce platform, providing relevant details such as shipping address and payment information.
2. After the order is confirmed, the customer service generates a unique order ID and associates it with the customer's purchase.
3. The customer service retrieves tracking information from the shipping carrier using the order ID.
4. The customer receives an order confirmation email or notification containing the order details and a link to track the order.
5. The customer clicks on the tracking link, which directs them to the order tracking page on the ecommerce platform.
6. On the order tracking page, the customer enters the order ID or any other required information to initiate the tracking process.
7. The e-commerce platform retrieves the latest tracking information from the shipping carrier's system.
8. The customer is presented with real-time updates on the order's status, including order processing, packaging, shipping, and estimated delivery dates.
9. The customer can view additional details, such as the shipping carrier's name, tracking number, and the order's current location if available.
10. The customer receives notifications or updates via email, SMS, or within the e-commerce platform regarding any changes or delays in the order's status.
11. Once the order is delivered, the tracking information is updated to reflect the successful delivery.
12. The customer can provide feedback, rate the delivery experience, or contact customer support through the e-commerce platform if needed.

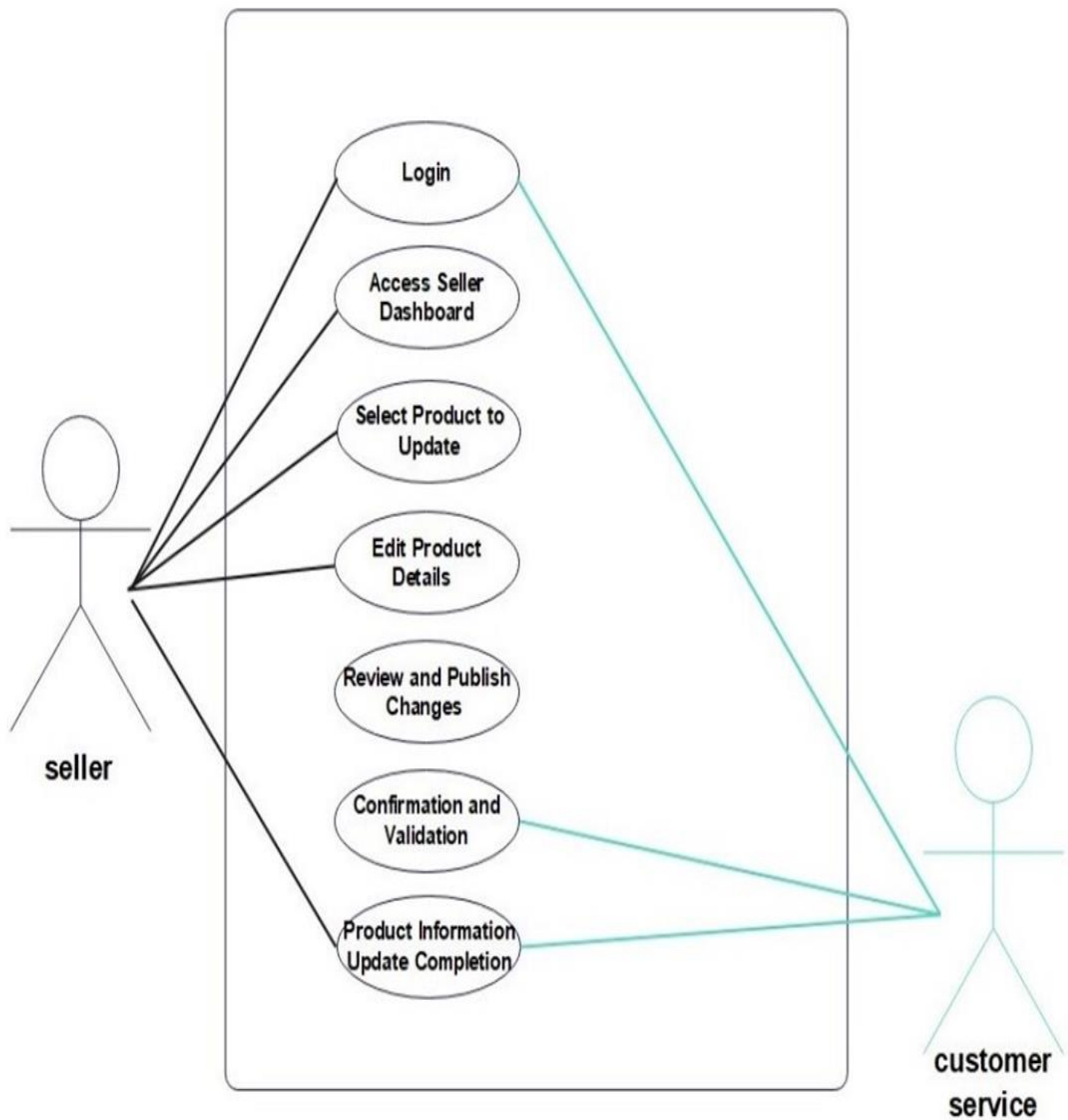
Alternative Sequence:

- If the customer enters an incorrect or invalid order ID, the system displays an error message and prompts them to enter the correct information.
- If there are delays or issues during the shipping process, such as customs clearance or weather conditions, the tracking information reflects these updates, and the customer is notified accordingly.
- If the order is canceled or returned, the tracking information is updated to reflect the change in status, and the customer is notified about the return process.

Postcondition:

1. The customer can track the progress and location of their order in real-time.
2. The e-commerce platform and shipping carrier provide accurate and up-to-date tracking information.
3. The customer receives timely notifications and updates regarding their order's status.
4. The customer's anxiety about the order's whereabouts is reduced, leading to improved customer satisfaction.

Scenario Product Information Use Case Model



Use Case Name: Product Information

Summary: The use case describes the process of Updating Product Information on an E-commerce Platform.

Actor:

1. Seller
2. customer service

Dependency: validate account use case.

Preconditions:

1. The seller has an active account on the e-commerce platform.
2. The seller has listed one or more devices for sale on the platform.

Main Sequence:

1. Access Seller Dashboard: The seller logs into their account on the e-commerce platform They navigate to the seller dashboard or seller management interface.
2. Locate Product Management: Within the seller dashboard, the seller identifies the section or option for managing product information They may find it labeled as "Manage Products," "Product Listings," or similar.
3. Select Product to Update: The seller reviews the list of their listed devices and selects the specific product they wish to update They may use search or filtering options to locate the desired product quickly.
4. Edit Product Details :The selected device's product page or editing interface is displayed The seller can modify various aspects of the product information, such as Title Update the product title or name Description: Edit the product description, highlighting key features, specifications, or other relevant details Images: Add, remove, or replace product images to showcase the device accurately Pricing: Adjust the product's price based on market conditions, promotions, or other factors Variations: If applicable, manage variations such as color, size, or storage options Inventory: Update the available quantity of the device in stock Shipping Options: Modify shipping details, including carrier options, costs, or delivery regions Other Attributes: Edit additional product attributes like brand, condition, warranty information, etc.
5. Save Changes: After making the necessary updates, the seller saves the changes to the product information Depending on the platform, this may involve clicking a "Save" or "Update" button or a similar action.
6. Review and Publish Changes: The seller has the opportunity to review the updated product information before making it live on the platform, They may preview the changes or consult a preview mode to ensure accuracy and quality Once satisfied, the seller selects the option to publish the updated product information.
7. Confirmation and Validation: e-commerce platform processes the updated product information and verifies its validity and compliance with platform policies Automated checks may be performed to ensure the accuracy and integrity of the information provided if any issues or discrepancies are identified, the seller may be alerted to make corrections.
8. Product Information Update Completion: The updated product information becomes live and visible to potential customers on the e-commerce platform Customers can now access the updated details, images, and pricing associated with the device.

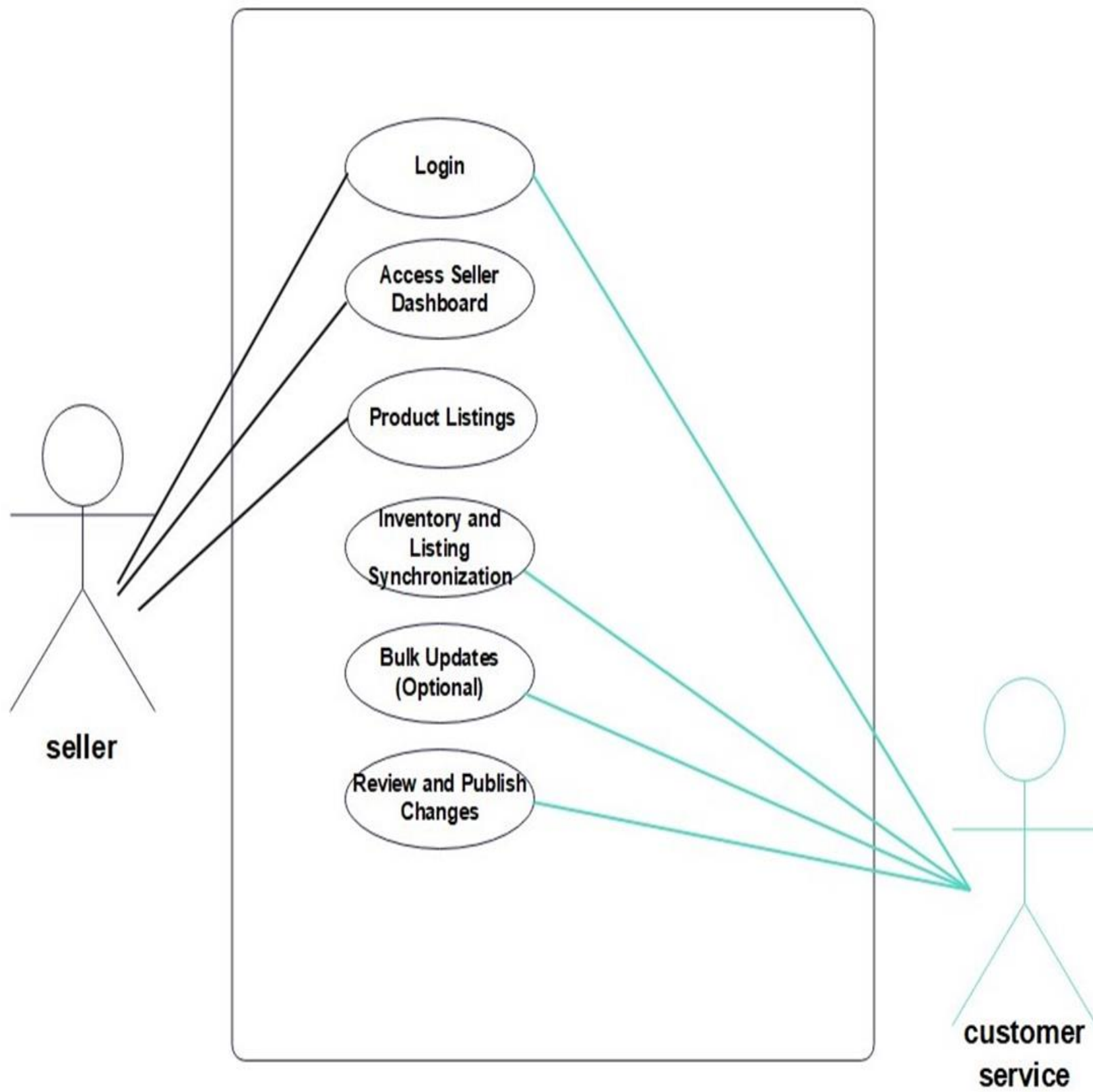
Alternative Sequence:

- If the seller encounters any issues or is unsure about specific aspects of the update process, they may seek support or guidance from the platform's customer service or help documentation.

Postconditions:

1. The product information for the selected device has been successfully updated on the e-commerce platform.
2. Customers can view and interact with the updated information when accessing the device's product page.

Scenario Manage Inventory and Listings Use Case Model



Use Case Name: Manage Inventory and Listings

Summary: The use case describes the process of Manage Inventory and Listings on an E-commerce Platform

Actor:

1. Seller
2. customer service.

Dependency: include validate account.

Preconditions:

1. The seller has an active account on the e-commerce platform.
2. The seller has listed products for sale on the platform.

Main Sequence:

1. Access Seller Dashboard: The seller logs into their account on the e-commerce platform They navigate to the seller dashboard or inventory management section.
2. Inventory Management: View Current Inventory: The seller can access a summary or detailed view of their current inventory, including the quantity and status of each product Stock Update: If necessary, the seller updates the inventory by adding new stock or adjusting quantities to reflect recent sales or replenishments Inventory Tracking: The platform may provide tools for tracking inventory movement, such as receiving stock transferring items between warehouses, or marking items as sold.
3. Product Listings View Existing Listings: The seller reviews their current product listings displayed on the platform Edit Listing Details: If changes are required, the seller selects the specific listing and accesses the editing interface Update Product Information: The seller can modify various aspects of the listing, including product title, description, images, pricing, variations, and other relevant attributes Listing Optimization: The seller ensures that the listing is optimized for search engines and potential customer searches, incorporating relevant keywords and improving visibility.
4. Inventory and Listing Synchronization Inventory Tracking: As the seller updates the inventory quantities or status, the platform synchronizes these changes across the live product listings Stock Availability: The platform ensures that the product listings accurately reflect the available stock, preventing overselling or displaying out-of-stock items Listing Updates: When the seller saves changes to the listing information, the platform updates the live product page with the modified details.
5. Bulk Updates (Optional) If supported by the e-commerce platform, the seller may perform bulk updates to inventory and listings This allows them to make changes to multiple products or listings simultaneously, saving time and effort.
6. Review and Publish Changes The seller reviews the updated inventory and listing information to ensure accuracy Once satisfied, they select the option to publish the changes, making them live on the e-commerce platform.

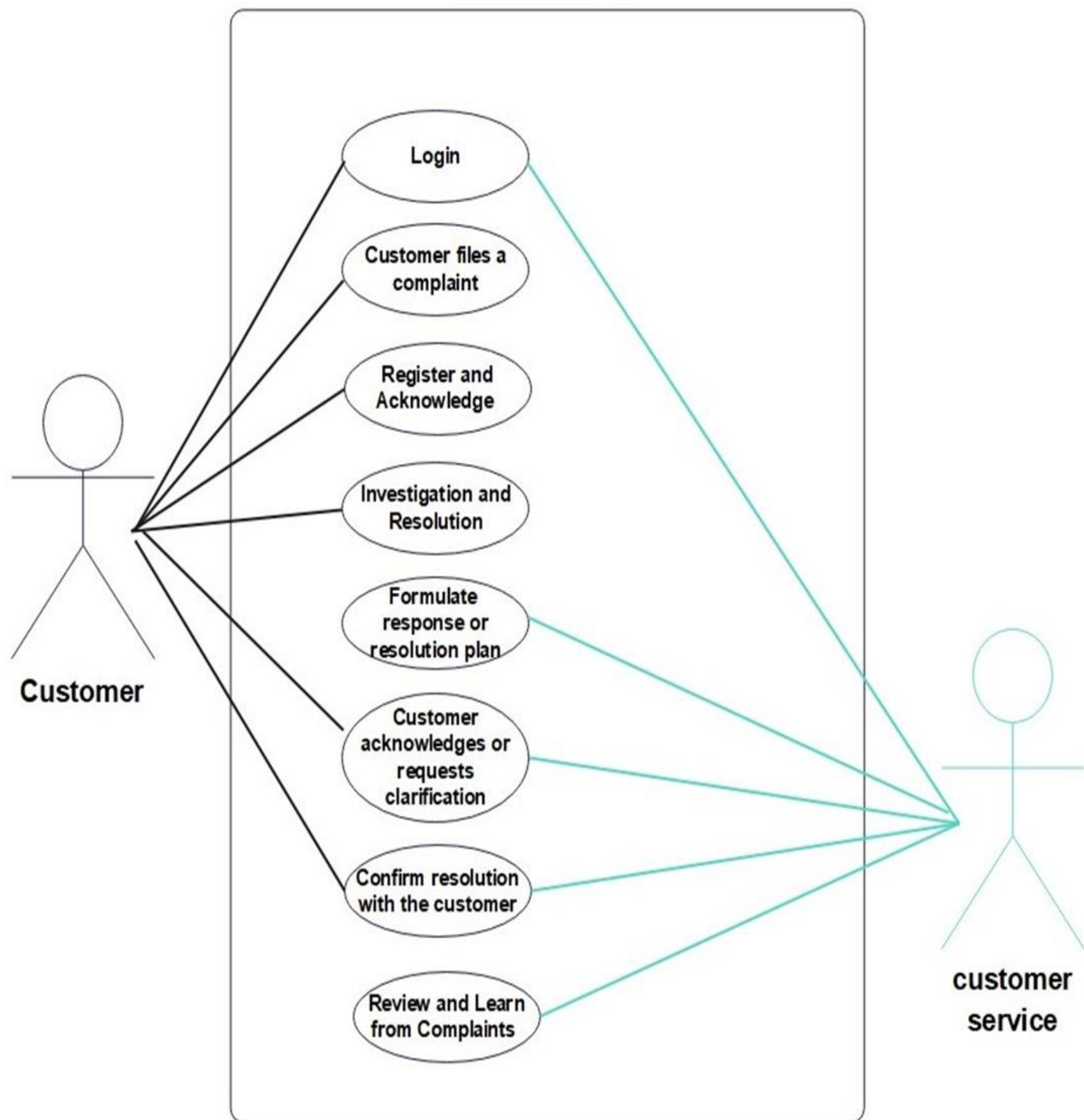
Alternative Sequence:

- If the seller encounters any issues or requires assistance, they may seek support from the platform's customer service or consult help documentation specific to inventory and listing management.

Postconditions:

1. The seller has successfully managed their inventory, ensuring accurate stock levels and availability.
2. The seller has updated and optimized their product listings to attract potential customers.
3. The inventory and listing changes are reflected live on the e-commerce platform, providing accurate information.to customers.

**Scenario complaints
Use Case Model**



Use Case name: complaints

Summary: The use case describes the process of complaints on an E-commerce Platform.

Dependency: Validate account use case.

Actors:

1. Customer.
2. Customer service.

Dependency: include validate account.

Preconditions:

1. The customer has experienced an issue or is dissatisfied with a product or service.
2. The organization has established a complaints management system or process.
3. The customer service is trained and equipped to handle customer complaints effectively.

Main Sequence:

1. The customer encounters an issue or is dissatisfied with a product or service and decides to file a complaint.
2. The customer contacts the organization through a preferred channel (e.g., phone, email, website form) to register their complaint.
3. The customer service receives the complaint and collects relevant information, such as the customer's details, description of the issue, and any supporting evidence.
4. The customer service acknowledges the complaint, either through an automated response or personal communication, to assure the customer that their complaint has been received.
5. The complaint is assigned a unique reference number or case ID for tracking and future reference.
6. The customer service investigates the complaint, gathering additional information if needed, and analyzing the root cause of the issue.
7. Based on the investigation, the customer support team formulates a response or resolution plan.
8. The customer service communicates the proposed resolution to the customer, providing clear explanations, options, and timelines for resolving the issue.
9. The customer acknowledges the proposed resolution, either accepting or requesting further clarification or adjustments if necessary.
10. If the proposed resolution is accepted, the customer support team proceeds with implementing the resolution plan.
11. Once the resolution is implemented, the customer support team confirms with the customer, ensuring that the complaint has been resolved to their satisfaction.
12. If the complaint is not resolved to the customer's satisfaction, the case may be escalated within the organization to higher-level management or specialized teams for further investigation and resolution.
13. The customer service reviews the overall complaints data periodically to identify recurring issues, trends, or areas for improvement in products, services, or processes.
14. Based on the insights gained from the complaints data, customer service takes appropriate actions to address systemic issues, improve customer satisfaction, and prevent similar complaints in the future.

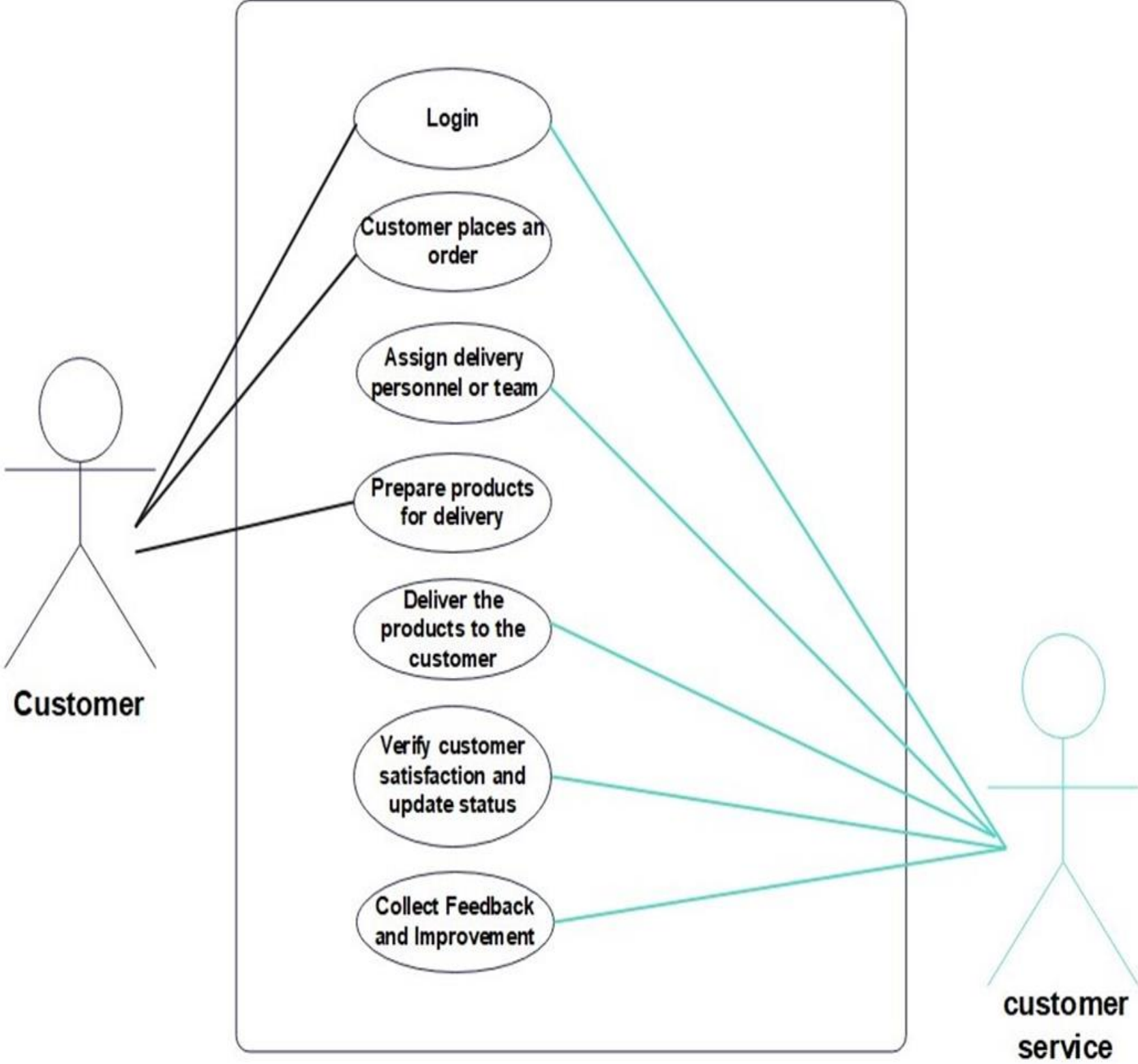
Alternative Sequence:

- If the initial complaint channel is unavailable or not responsive, the customer may use alternative channels (e.g., social media, in-person) to register their complaint.
- In cases where the complaint involves legal or compliance issues, the organization may involve legal or compliance experts to ensure proper resolution.
- If the complaint involves a serious or urgent matter, the organization may prioritize the investigation and resolution, providing expedited handling and communication to the customer.
- If the customer is dissatisfied with the resolution proposed by the customer service, they may request to speak with a supervisor or escalate the complaint to a higher level within the organization.

Postcondition:

1. The customer's complaint is addressed and resolved in a satisfactory manner.
2. The organization maintains a record of the complaint, including details of the issue, investigation, resolution, and customer feedback.
3. The organization continuously improves its products, services, or processes based on insights gained from customer complaints.
4. The customer's trust and satisfaction with the organization are restored or enhanced, leading to improved customer loyalty and retention.

Scenario Delivery
Use Case Model



Use Case name: Delivery

Summary: The use case describes the process of Delivery on an E-commerce Platform.

Actors:

1. Customer
2. Customer service

Preconditions:

3. The organization has received an order from the customer.
4. The organization has established a delivery management system or process.

Main Sequence:

1. The customer places an order through the organization's preferred channel (e.g., website, mobile app, phone).
2. The organization receives and processes the order, including verifying payment and preparing the product or service for delivery.
3. The customer service receives the order details and assigns a delivery personnel or team based on factors such as proximity, availability, and delivery requirements.
4. The customer service receive the necessary information and instructions for the delivery, including the customer's address, contact details, and any specific delivery preferences or instructions.
5. The delivery personnel prepare the products or services for transportation, ensuring proper packaging and handling to prevent damage or loss.
6. The delivery personnel navigate to the customer's location using the most efficient route, considering factors such as traffic conditions and delivery time windows.
7. The delivery personnel arrive at the customer's location and verify the recipient's identity, ensuring that the delivery is made to the correct person.
8. The delivery personnel hand over the products or provide the services to the customer, ensuring that they are in good condition and meeting the customer's expectations.
9. The customer inspects the delivered products or services and confirms their satisfaction.
10. If applicable, the customer signs a delivery confirmation or provides an electronic acknowledgement of the successful delivery.
11. The delivery personnel update the delivery status in the system, indicating that the delivery has been completed.
12. The organization may collect customer feedback on the delivery experience, product quality, or service level to improve future deliveries.
13. In cases where the customer is not available to receive the delivery, the delivery personnel follow predefined protocols, such as attempting redelivery, leaving a delivery notification, or contacting the customer to reschedule.
14. The organization's delivery management system updates the order status and provides tracking information to the customer, enabling them to monitor the progress of their delivery if available.

Alternative Sequence:

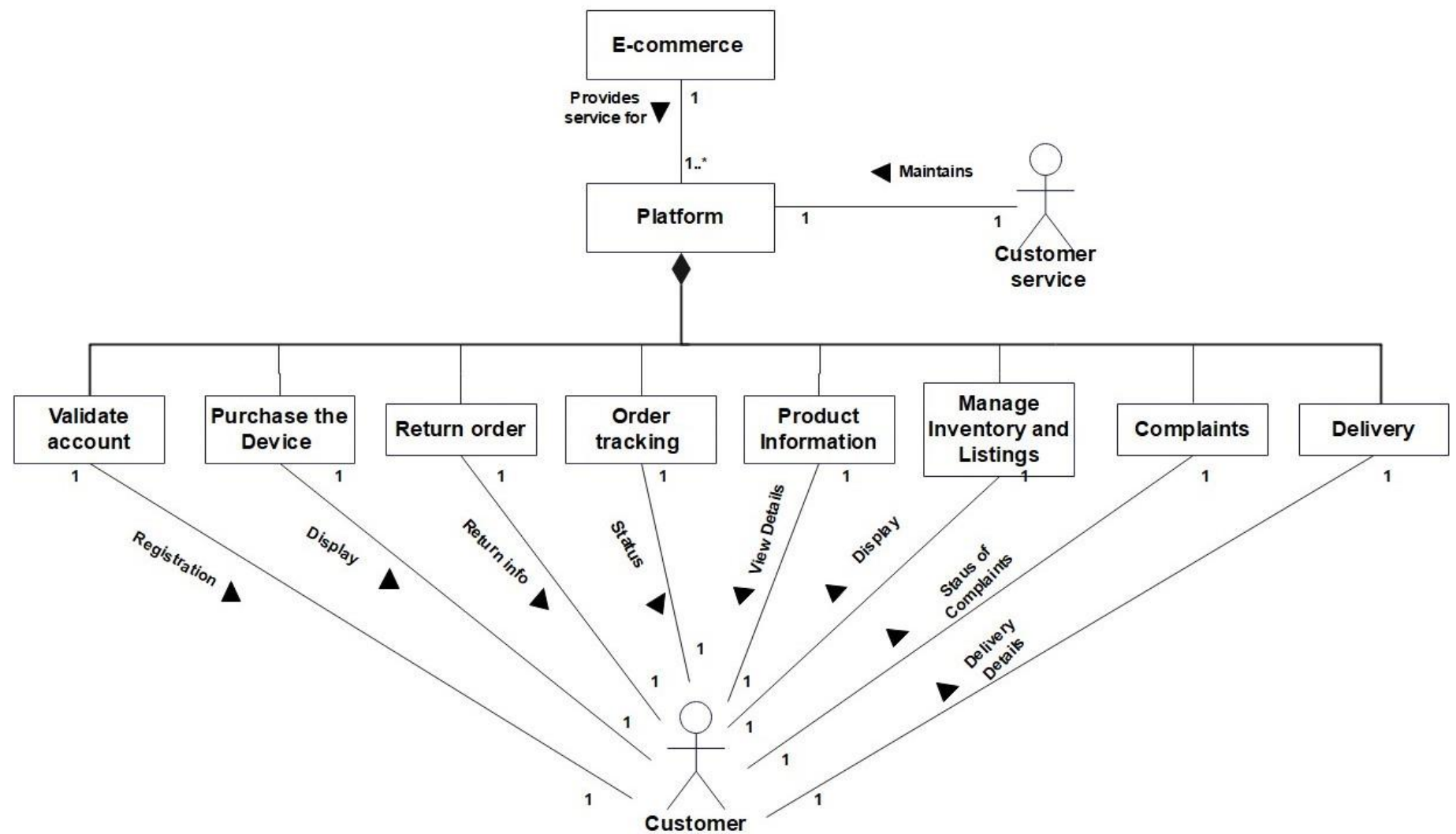
- If the customer is not available at the given address during the delivery attempt, the delivery personnel may try.
- .to contact the customer for an alternative delivery arrangement.
- If the customer cannot be reached or an alternative arrangement cannot be made, the delivery personnel may leave.
- a delivery notification or attempt delivery on another day
- If multiple delivery attempts fail, the shipment may be returned to the e-commerce platform or held at a local pickup location for the customer to collect.

Postcondition:

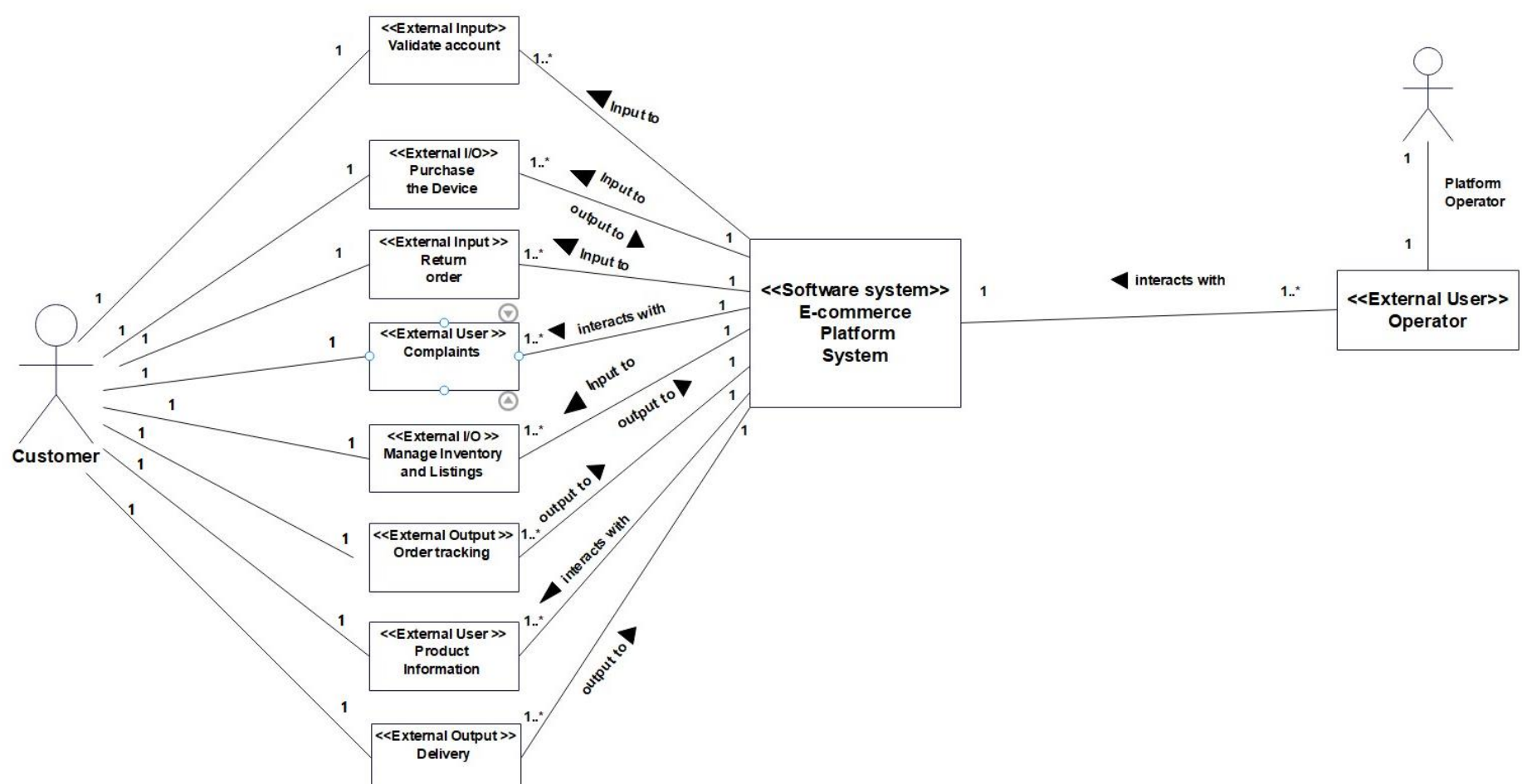
1. The customer receives the products or services in a timely and satisfactory manner.
2. The delivery personnel complete the delivery process efficiently, ensuring minimal delays or issues.
3. The organization maintains accurate delivery records and tracking information.
4. The customer's confidence in the organization's delivery capabilities is reinforced, leading to improved customer satisfaction and loyalty.

Stage 2: Develop Software Analysis Model - Static Model. (week 5).

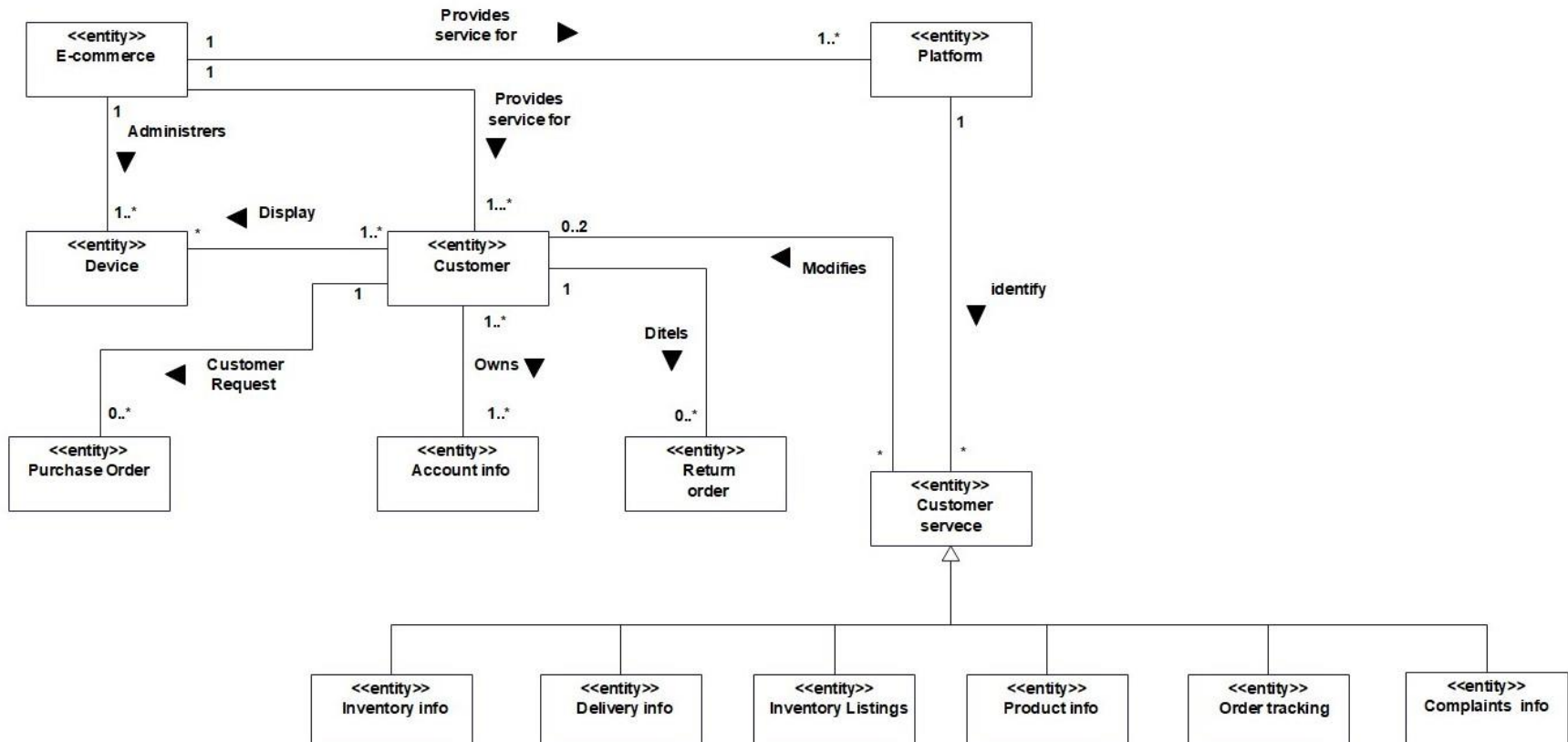
a. Draw the Conceptual Static Model for your project.



b. Draw the Context Class Diagram for your project.



c. Draw the Entity Class Model for your project and draw the class attributes for each entity



<<entity>> E-commerce
EcommerceID=int Platform =string CustomerService=string Product =string

<<entity>> Platform
PlatformID =int URL =string Name =String HostingDetails =String SecurityFeatures =String

<<entity>> Customer
CustomerID = int Name =String Email =String Password =String

<<entity>> Device
DeviceID = int Name =String Description =String Price =Double

<<entity>> Account info
AccountID = int CustomerID =int Type =String Status =String

<<entity>> Purchase Order
OrderID = int CustomerID =int DeviceID =int Quantity =String Date =String

<<entity>> Return order
ReturnID = int OrderID =int Reason =String

<<entity>> Customer service
ServiceID = int ContactChannels =String ResponseTime =String ServiceQuality =String CustomerFeedback=String ServiceQuality =String

<<entity>> Order tracking
TrackingID = int OrderID=int Status =String

<<entity>> Inventory
InventoryID = int CustomerID =String ProductID =String QuantityAvailable =String Location =String

<<entity>> Complaints
ComplaintID = int CustomerID =int Issue =String ResolutionStatus =String

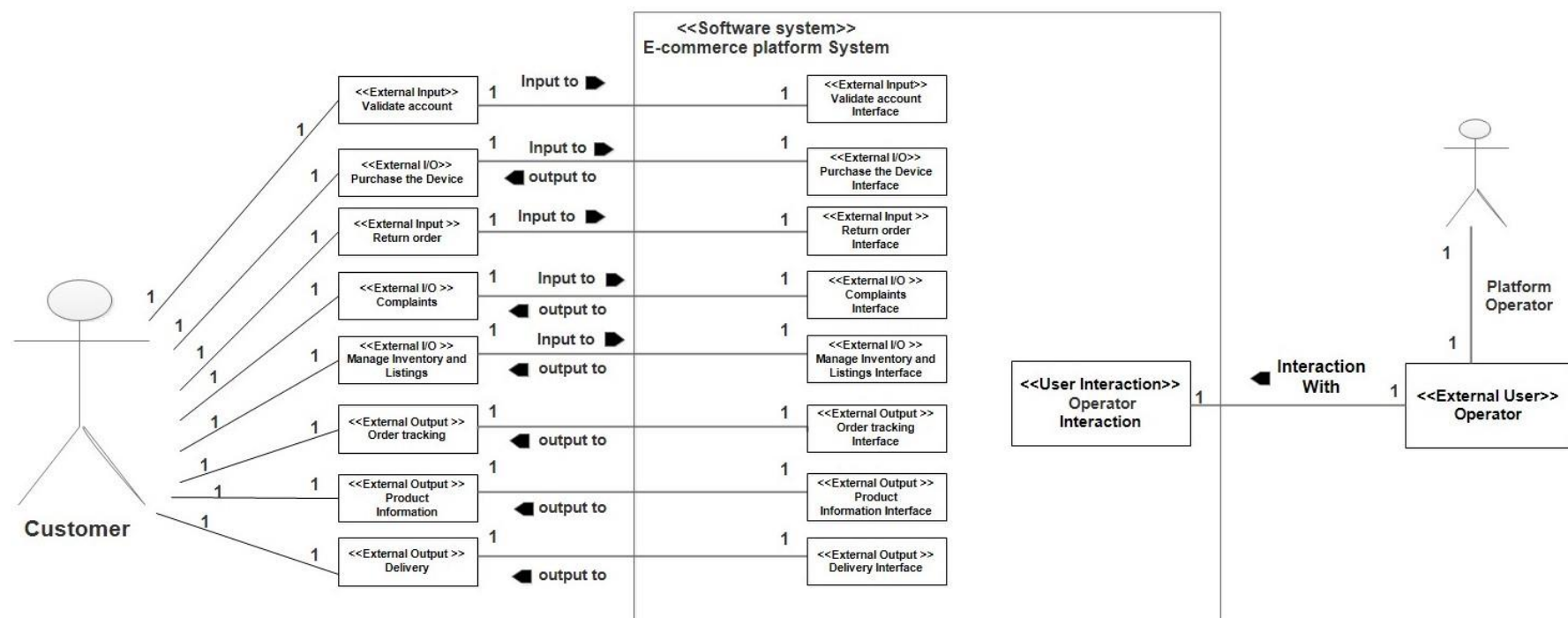
<<entity>> Delivery
DeliveryID = int OrderID =int DeliveryDate =String Delivery Status =string

<<entity>> Product info
ProductID = int DeviceID =int Price =Double Specifications =String

<<entity>> Inventory Listings
ListingID = int ProductID =String StockQuantity =int Price =Double

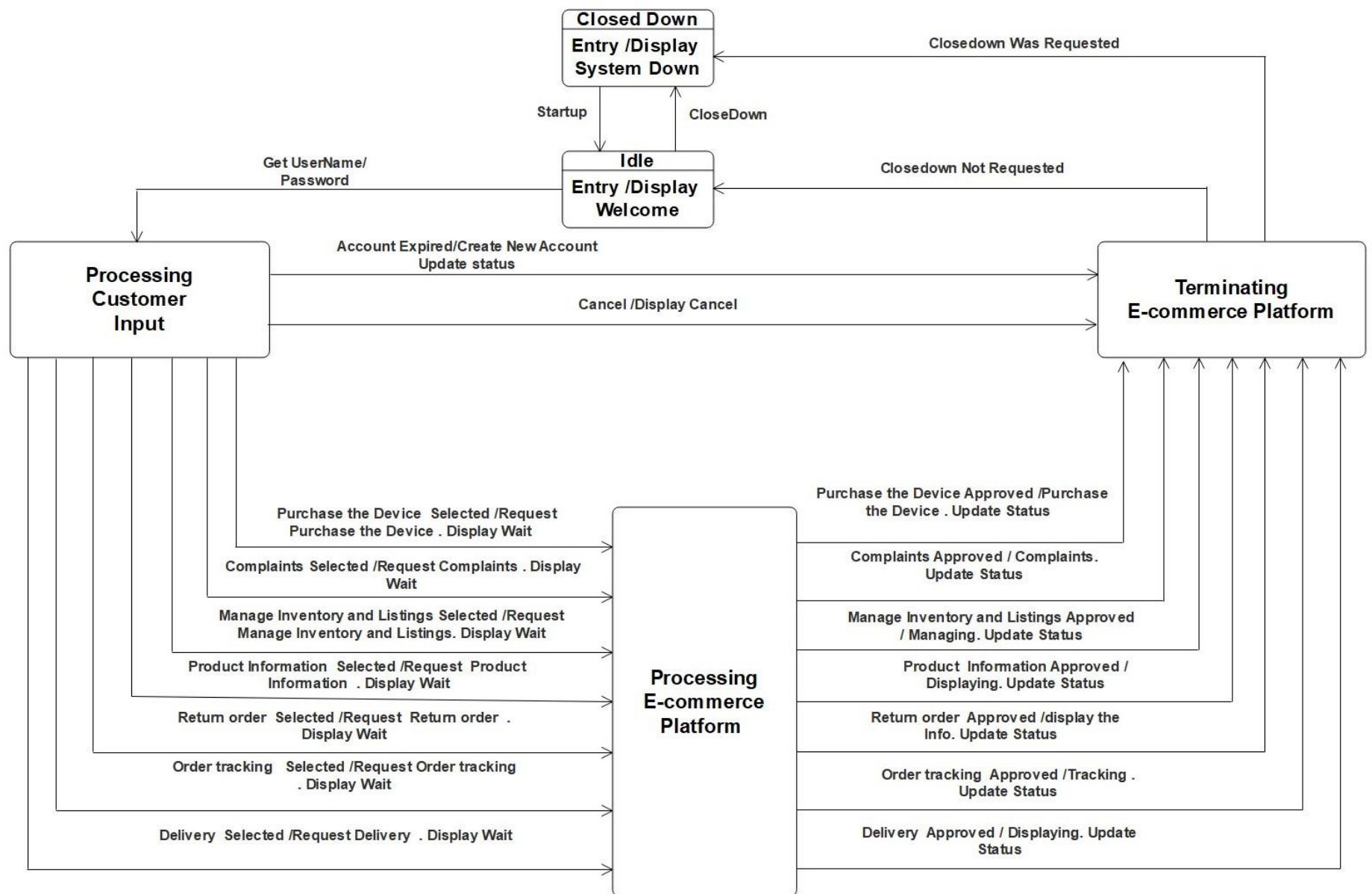
Stage 3: Develop Software Analysis Model – Structure
system into objects. (week 6).

a. Expand the context class diagram to show the boundary classes that communicate with the external classes.

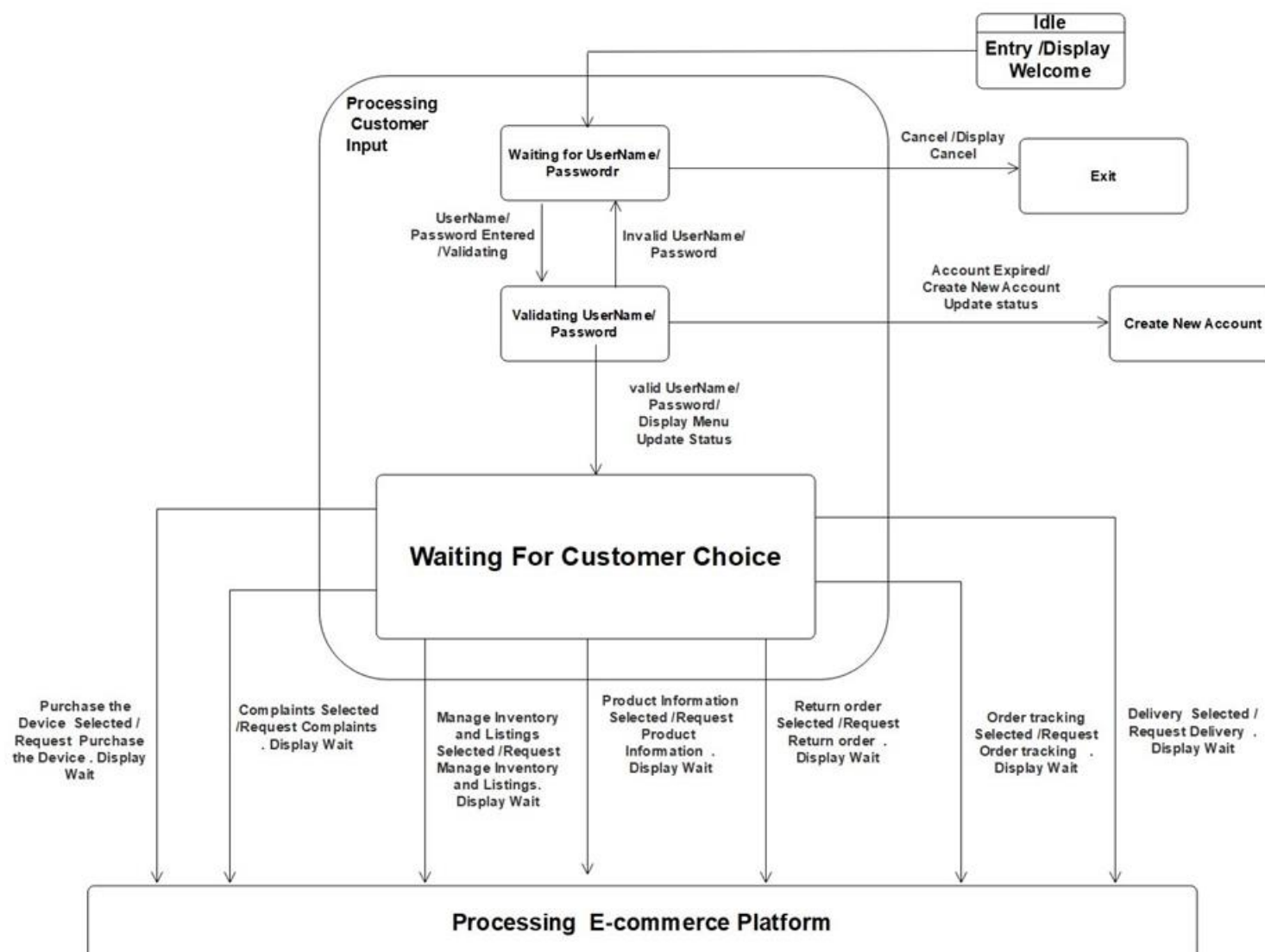


Stage 4: Develop Software Analysis Model – Develop state charts for state dependent objects. (week 7).

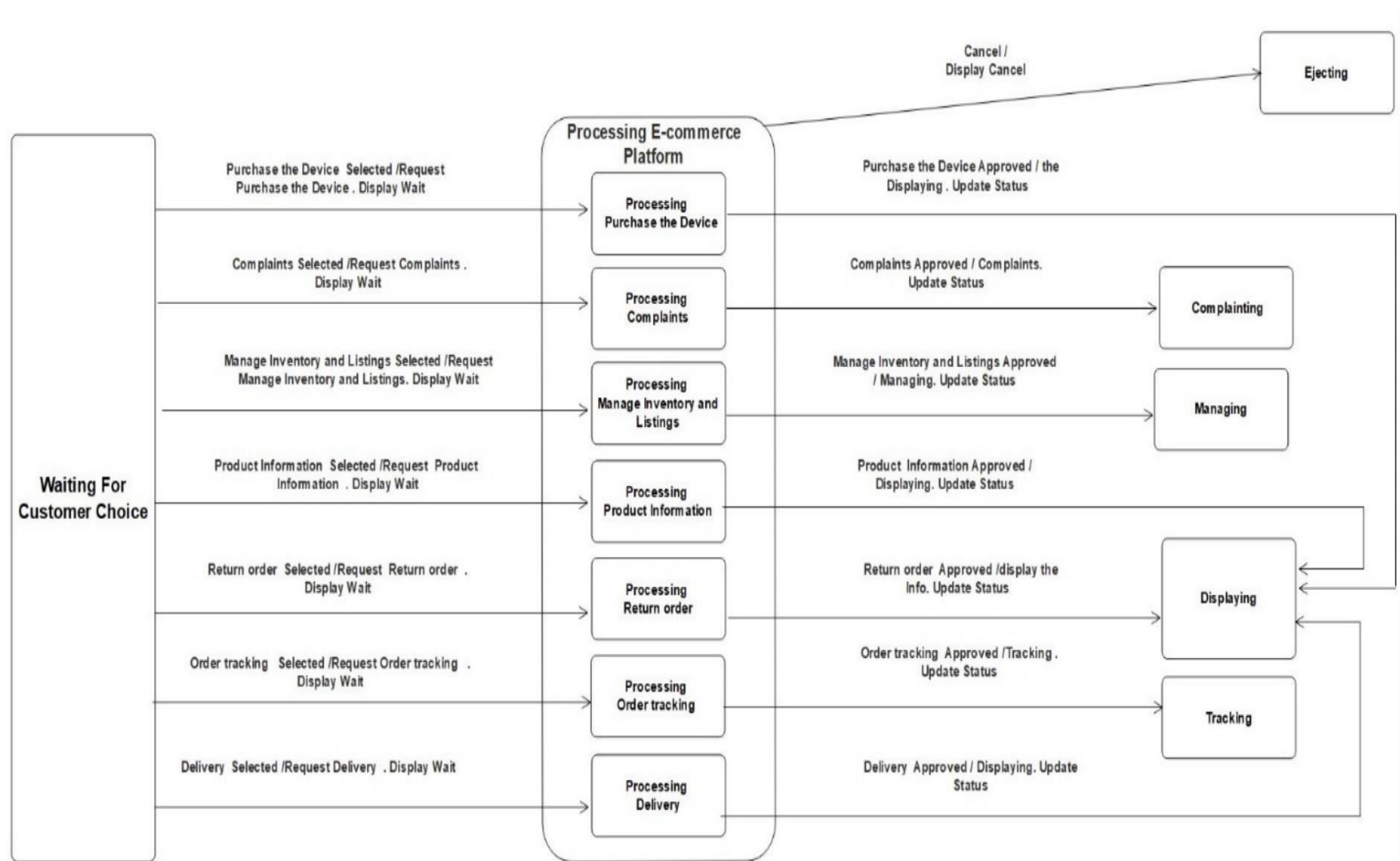
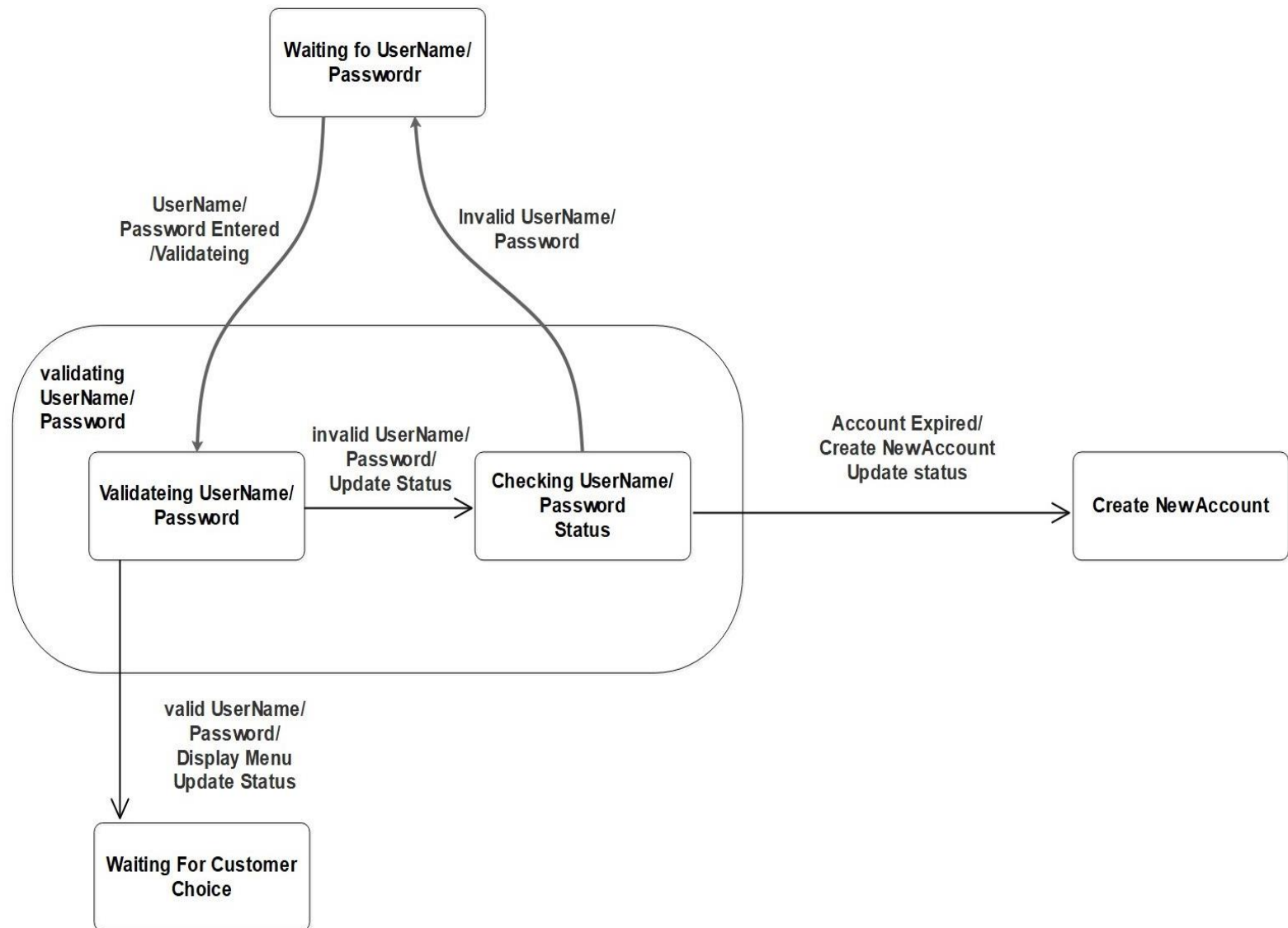
a. Draw top level state chart for your project based on the use case.

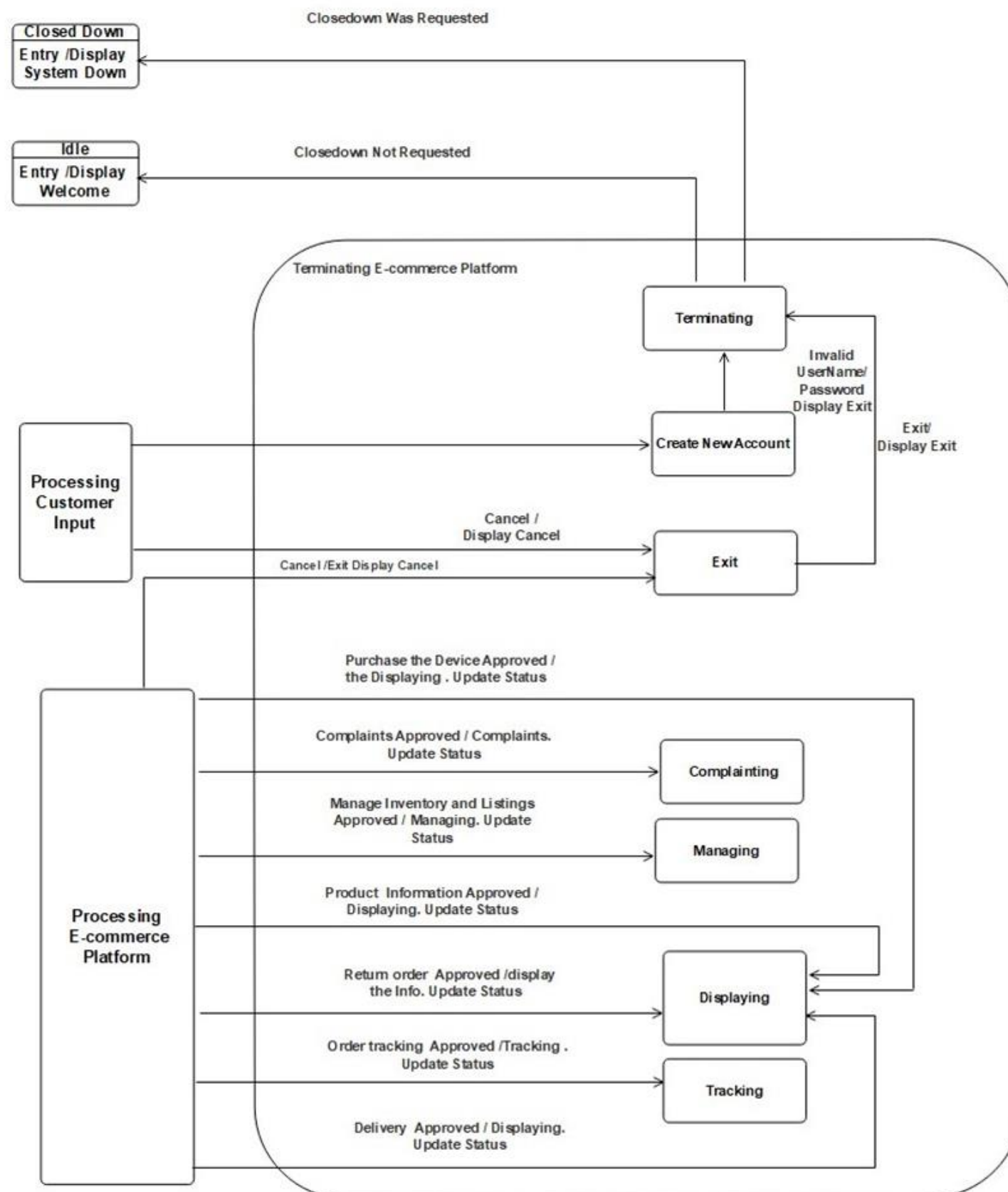


b. Decompose each composite state in a separate chart by representing the underlying substates of each one.



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Stage 5: Develop Software Analysis Model – Develop object interaction diagrams for each use case. (week 8).

a. Draw either the communication diagram or sequence diagrams for each use case.

